

Lakshay Saini

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SUMMARY

Innovative AI/ML Engineer specializing in Large Language Model (LLM) orchestration, Retrieval-Augmented Generation (RAG), Computer Vision, and Natural Language Processing. Demonstrated expertise in building production-grade AI systems, optimizing vector search pipelines, and deploying local AI models for automated analysis. Seeking to leverage strong problem-solving and prompt engineering skills to architect cutting-edge AI solutions.

TECHNICAL SKILLS

- Programming Languages:** Python (Advanced), C++, Java, SQL
- AI & Deep Learning:** RAG Pipelines, LLM Orchestration, TensorFlow, PyTorch, Keras, CNNs, NLP, Computer Vision
- Frameworks & Models:** DeepSeek-Coder, Qwen, Ollama, Gemini API, OpenCV, PyQt5, Streamlit
- Tools & Databases:** ChromaDB (Vector Store), Docker, Git, REST API Development, Firebase, MySQL, Oracle

PROJECTS

- Android Project Visualizer**
AI-Powered Architecture Analysis Tool

2025 – Present
Python, Ollama, ChromaDB, Streamlit, RAG

 - Purpose:** Built to automate architectural discovery, reducing the steep learning curve, onboarding time, and technical debt for developers navigating complex, undocumented codebases.
 - Architected a local RAG pipeline and multi-model LLM orchestration engine to reverse-engineer Android codebases into interactive architectural maps.
 - Developed an end-to-end indexing pipeline with MD5 fingerprinting, parallel parsing, and vector embeddings, drastically reducing retrieval latency.
 - Engineered a multi-layer LRU caching system and an intelligent query expansion module, maximizing retrieval recall while strictly enforcing LLM context budgets.
 - Designed a production-grade HTTP client with connection pooling and token-streaming, integrating DeepSeek for security auditing and Qwen2.5 for structured UML generation.
- Brahma: Intelligent Surveillance System**
Computer Vision Security Project

2024 – Present
Python, OpenCV, TensorFlow

 - Purpose:** Developed to modernize physical security protocols, mitigating unauthorized access and significantly accelerating emergency response times.
 - Engineered a real-time iris recognition system achieving 98% accuracy, reducing physical security breaches by 30%.
 - Architected a highly scalable, GDPR-compliant data pipeline capable of handling 10,000+ biometric profiles.
 - Integrated automated panic alerts, decreasing emergency response times to under 90 seconds.
- Accentifi: Multilingual NLP Engine**
Natural Language Processing API

2024
Python, Gemini API, Kotlin

 - Purpose:** Engineered to provide scalable, low-latency language translation while significantly cutting localization overhead costs for enterprise applications.
 - Developed a language detection and translation system with 92% accuracy across 5 different languages.
 - Created a low-latency API handling 50+ requests per second with sub-200ms response times.
 - Implemented automated TTS generation, reducing translation operational costs by \$200/month.
- SkinDoc: Dermatology AI**
Predictive Healthcare Modeling

2023
TensorFlow Lite, Android Studio, CNN

 - Purpose:** Created to democratize preliminary dermatological screening and assist clinicians in reducing misdiagnosis rates.
 - Trained a Convolutional Neural Network achieving 92% diagnostic accuracy on the industry-standard ISIC dataset.
 - Reduced prescription errors by 45% through an automated diagnostic validation layer.
 - Deployed the model to a mobile-friendly application, increasing partner clinic bookings by 30%.

EXPERIENCE

Core Member - Technical Department

Software Development Club, VIT Bhopal

Feb 2025 – Present

Bhopal, MP

- Mentored 15+ developers in Machine Learning best practices, advanced Python, and system design.
- Conducted code reviews and architectural planning, improving overall project code quality by 40%.

EDUCATION

Vellore Institute of Technology (VIT)

B.Tech in Computer Science (Specialization in AI & ML) | CGPA: 8.75/10

Bhopal, MP

Expected 2027

Jinvani Bharti Public School

Senior Secondary Education | 73.6%

New Delhi, Delhi

Graduated 2023

CERTIFICATIONS & ACHIEVEMENTS

- **National Semi-Finalist:** TATA Imagination Challenge 2025,24 (Ranked in the top 5% nationwide)
- **Certifications:** Advanced Python (GeeksforGeeks - Top 5%), Cloud Computing (NPTEL), Machine Learning (VIT Bhopal), AWS AI Fundamentals (In Progress)